

XIMO

Technology Stack

A concise map of the client, backend, data layer, integrations, and delivery tooling behind the Ximo app.

REPOSITORY SNAPSHOT · 27 SEPTEMBER 2026

PRODUCT SURFACES

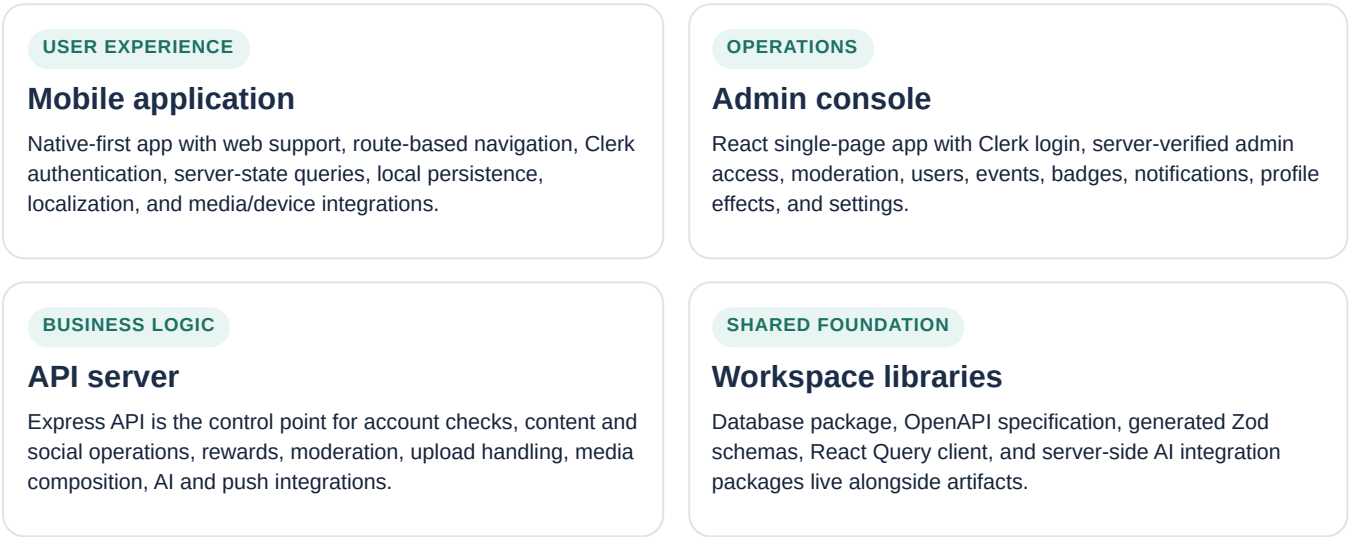
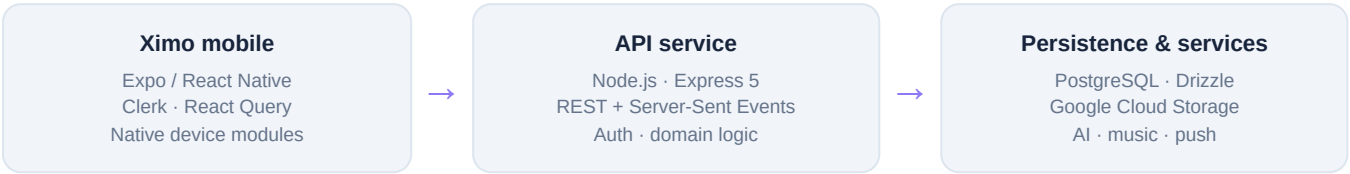
Expo mobile app · Express API · PostgreSQL data layer ·
React/Vite admin console

ARCHITECTURE SHAPE

TypeScript pnpm monorepo with shared API contracts
and an API-centered service boundary.

Four layers, shared contracts

Ximo is organized as a pnpm workspace: the mobile client and admin console call a central API, while shared libraries provide database schemas, generated validation, and typed API utilities.



Core pattern: the client renders and caches the experience; authenticated API routes enforce identity and application rules; PostgreSQL stores structured state; object storage holds media. Some local cache and retry patterns improve resilience but do not constitute a full offline mode.

Data path examples

Read: Ximo obtains a Clerk session → API validates the session and user scope → Drizzle queries PostgreSQL → client updates query/context state.

Media: app uploads image/video → server checks limits and media properties → GCS stores assets → API returns a media reference used by content features.

Real time: authenticated SSE streams carry selected events; client reconnect logic and polling support foreground refresh. Current event fanout is process-local.

Frameworks and responsibilities

Version numbers below are declared package versions or shared catalog ranges. The committed pnpm lockfile resolves the exact installed package graph.

LAYER	TECHNOLOGY	ROLE IN XIMO
Mobile	Expo SDK ~54.0.37 React Native 0.81.5 React 19.1.0 TypeScript ~5.9.2	Cross-platform client for iOS, Android and web preview; typed UI and application logic.
Navigation & data	Expo Router ~6.0.24 TanStack React Query 5.x Zod 3.25.76 AsyncStorage 2.2.0	File-based routing; API query/cache state; runtime validation at selected boundaries; local persistence and cached-first UX.
Mobile identity / native	Clerk Expo ~3.2.7 Expo Notifications / Secure Store / Image Picker / Video / Audio Reanimated 4.x	Authentication, push, secure token cache, image/video/audio/device features and UI animation.
API service	Node.js · Express 5 TypeScript · Zod Clerk Express ~2.1.12	REST endpoints, auth middleware, validation, content/social/reward rules, admin APIs and SSE stream endpoints.
Database	PostgreSQL Drizzle ORM ~0.45.2 node-postgres (pg) ~8.20.0	Relational persistence for profiles, journals, stories, social graph, messages, moderation, guide sessions, rewards and creator publishing.
Admin web	React 19.1.0 · Vite 7.3.x Tailwind CSS 4.1.x · Radix UI TanStack Query · Recharts Clerk React ~5.61.9	Browser-based moderation and product-operations console; component primitives, forms, charts and server data access.
Shared contracts	OpenAPI 3.1 Generated API client Generated Zod schemas	Reusable request/response types, client hooks and selected runtime schemas across app and API.

Mobile interface libraries

React Navigation

Gesture Handler

Safe Area Context

SVG

i18next

react-i18next

Expo Location

Expo Sharing

API security / observability

Helmet

CORS

express-rate-limit

Pino

Pino HTTP

Supertest

Semver note: values shown with ~ or ^ are manifest ranges, not immutable exact runtime versions. For reproducible build records, use the lockfile and capture the commit/build artifact used for each release.

Infrastructure and operating tools

SERVICE / TOOL	TECHNOLOGY	USE
Authentication	Clerk	Mobile and admin sign-in; server middleware validates user identity and admin privilege.
Media storage	Google Cloud Storage	Stores image, video and generated media objects; served through API media routes.
Image processing	Sharp	Server-side image conversion/validation and optimization.
Video composition	ffmpeg / ffprobe	Server-side video inspection and composition; executable availability is part of deployment requirements.
Generative AI	Anthropic SDK OpenAI-compatible image integration	Story assistance, drift/admin generation, and manga/story image generation.
Music	Audius API	Track search and stable music stream metadata for stories and creative features.
Push	Expo Push API	Device token registration and notification delivery.
Build / preview	EAS Build · Expo CLI Replit workflows	Internal and production mobile build profiles; workspace-managed development workflows.

Developer quality tools

- **Monorepo manager:** pnpm 10.26.1 with shared dependency catalog and committed lockfile.
- **Type system:** TypeScript ~5.9.2; root scripts typecheck shared libraries and artifact packages.
- **Lint / format:** ESLint 10.x and Prettier 3.x.
- **Tests:** Vitest 4.x; API tests use Supertest where appropriate.
- **Build:** Expo/EAS for mobile; Vite for admin; esbuild for API packaging.

Build configuration

- EAS profiles include internal Android APK and production Android App Bundle / iOS device builds.
- Build profiles declare Node 22 and pnpm 10.26.1 for mobile builds.
- Expo OTA updates are disabled, so app-level fixes require a new native build / distribution cycle.
- Only Android EAS workflow files were found in GitHub Actions; workflows are manually triggered.

Operational characteristics to know

API is a single Express service backed by PostgreSQL and GCS. Current SSE fanout and several throttles/counters are process-local; choose a single-instance limit or add shared coordination before horizontal scaling. Startup migrations run after the server opens its port. Health response is process-level and does not establish database or provider readiness.

Contract / privacy note: shared generated API contracts cover only part of the route surface. Public media endpoints and intended-private journal/gallery data should be checked together against the storage visibility policy. These are implementation notes, not claims that a production incident has occurred.

Source map

Package manifests: artifacts/sky-journal/package.json; artifacts/api-server/package.json; artifacts/admin/package.json; lib/db/package.json. Workspace: package.json; pnpm-workspace.yaml; pnpm-lock.yaml. Runtime/config: artifacts/sky-journal/app.json; eas.json; artifacts/api-server/src/app.ts; src/index.ts; src/routes/index.ts; src/routes/health.ts; lib/db/src/index.ts; lib/api-spec/openapi.yaml. Integrations: artifacts/api-server/src/routes/upload.ts; routes/media.ts; routes/music.ts; routes/manga-generation.ts; services/pushService.ts; shared OpenAI integration package.

Prepared 27 September 2026 · Describes repository-declared stack and architecture; does not certify deployed configuration or provider availability.